

Pre-Course Assessment

Complete before Module 0. Answer without using the guide or source code.

Student: _____

Date: _____

Pathway: Code Learning / Robot Programming

Team: _____

Scoring: Questions 1-6 are worth 1 point each. Questions 7-12 are worth 2 points each. Total: 18 points.

1. What is the difference between `int`, `double`, and `bool`?

2. What does `==` mean, and how is it different from `=`?

3. What does a `while` loop do?

4. What is the purpose of a header file?

5. Where are VEX motors and sensors configured?

6. What does the heading argument in `move( )` control?

7. Why must the software gear ratio match the physical drivetrain?

8. What does the proportional term of PID respond to?

9. What is integral windup?

10. Why should an autonomous routine be tested multiple times?

11. Name two non-software factors that can change autonomous performance.

12. When should a team consider moving from VEXLearn to an odometry-based template?

Score Record

Initial score: _____ / 18 Confidence before the course (1-5): _____